

EMD NEDUET
EC-101 Microeconomics
Production Possibility Curve (PPC)

1. What is a Production Possibility Curve (PPC)? Explain the relationship between scarcity, choice, and opportunity cost using the PPC framework. Why does a PPC normally slope downward?
2. Consider an economy producing only consumer goods and capital goods. Explain how the following situations would affect the PPC:
 - a) An increase in the economy's labor force
 - b) Improvement in production technology
 - c) A natural disaster that destroys productive resources
 - d) An increase in the quantity of capital

For each case, state whether the PPC shifts outward, shifts inward, or remains unchanged, and explain why.

3. An economy can produce either food (F) or clothing (C). Its production possibilities are:

Combination	Food (units)	Clothing (units)
A	0	50
B	10	45
C	20	38
D	30	28
E	40	15
F	50	0

- a) Calculate the opportunity cost of producing each additional 10 units of food.
 - b) Is the opportunity cost constant or increasing?
 - c) What does your answer suggest about the shape of the PPC?
 - d) Which economic concept explains the increasing opportunity cost?
 - e)
4. Differentiate between points on, inside, and outside the PPC. **Use a clearly labelled PPC diagram to support your answer.** Explain what each position represents in terms of:
 - Productive efficiency

- Underutilization of resources
 - Scarcity/infeasibility
5. Distinguish between a movement along the PPC and a shift of the PPC. Give one example of each.
 6. How can technological progress affect a Production Possibility Curve? Is it possible for technological progress to increase the economy's ability to produce one good without increasing its ability to produce another? Explain with a diagram.
 7. An economy has two alternatives: producing food or producing industrial machinery. If the government allocates more resources to machinery production today, what is the likely opportunity cost? How might this decision affect the economy's PPC in the future?
 8. Is a point outside the PPC necessarily undesirable? Explain why such a point is currently unattainable but may become attainable in the future.
 9. Consider the following PPC:

Combination	Good X	Good Y
A	0	60
B	10	55
C	20	47
D	30	36
E	40	22
F	50	0

An economy is currently producing 30 units of X and 36 units of Y.

- a) Is this combination efficient?
- b) What is the opportunity cost of increasing X from 30 to 40 units?
- c) If the economy produces 40 units of X and only 15 units of Y, is the economy operating on, inside, or outside the PPC? Explain.
- d) If it wants to produce 50 units of X and 20 units of Y, is this combination attainable? Why?